

Overview of the Immune System

Danger sensing, innate vs adaptive organization, inappropriate responses, and resolution

If you remember nothing else

- The immune system protects against danger, but disease can also result when immune responses are inappropriate.
- Immune responses are organized into innate and adaptive arms with different receptor strategies, timing, and memory.
- The lecture frames immune activation around detection of danger signals rather than a simple self-versus-nonself rule.
- After the threat is controlled, the response must contract and transition toward cleanup, repair, and memory.

Core framework

- Innate immunity: rapid pattern-based recognition using germline-encoded receptors.
- Adaptive immunity: antigen-specific B- and T-cell responses built around rearranged antigen receptors.
- The two systems are linked rather than isolated: innate sensing helps initiate and shape adaptive responses.

Recognition anchors

- Ask first: Is the vignette testing rapid pattern recognition or antigen-specific memory?
- Harmless nonself material can be tolerated, while altered self can trigger immunity; context matters.
- Resolution is an active end-phase of an immune response, not simply the absence of inflammation.

Fast contrasts

- Innate = rapid + recurring danger patterns.
- Adaptive = rearranged receptors + specificity + memory.
- Excessive/misdirected immunity can itself become a cause of clinical disease.